

5. (a) (i) Draw a dot and cross diagram to show the bonding in a carbon dioxide molecule. [2]

carbon (2,4)

oxygen (2,6)

- (ii) Explain why carbon dioxide has a low boiling point. [2]

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- (b) (i) Draw a diagram to show the electronic changes that take place during the formation of magnesium oxide. Include the charges on the ions formed. [2]

magnesium (2,8,2)

oxygen (2,6)

- (ii) Explain why magnesium oxide has a higher melting point than sodium chloride. [2]

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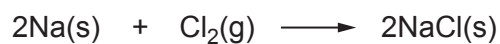
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- (iii) Sodium chloride is produced by the reaction of sodium with chlorine.



In a reaction using 0.080 mol of sodium, 4.12 g of sodium chloride was produced.

Calculate the percentage yield of this reaction. [3]

$$A_r(\text{Na}) = 23$$

$$A_r(\text{Cl}) = 35.5$$

Percentage yield = %

